

```
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ nano command_executor.c
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ gcc command_executor.c -o command_executor
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ ls
command_executor  command_executor.c
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ ./command_executor

Enter a Linux command (or type 'exit' to quit): ls

----- Parent Process -----
Parent PID : 963
Child PID  : 964

----- Child Process -----
Child PID  : 964
Parent PID : 963
Executing command: ls

command_executor  command_executor.c

Command 'ls' executed successfully.

Enter a Linux command (or type 'exit' to quit): date

----- Parent Process -----
Parent PID : 963
Child PID  : 976

----- Child Process -----
Child PID  : 976
Parent PID : 963
Executing command: date

Tue Aug 11 03:04:21 UTC 2026

Command 'date' executed successfully.

Enter a Linux command (or type 'exit' to quit): whoami

----- Parent Process -----
Parent PID : 963
Child PID  : 977

----- Child Process -----
Child PID  : 977
Parent PID : 963
Executing command: whoami
```

```
bharath

Command 'whoami' executed successfully.

Enter a Linux command (or type 'exit' to quit): uname

----- Parent Process -----
Parent PID : 963
Child PID  : 978

----- Child Process -----
Child PID  : 978
Parent PID : 963
Executing command: uname

Linux

Command 'uname' executed successfully.

Enter a Linux command (or type 'exit' to quit): exit
Exiting program...
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ nano file_copy.c
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ gcc file_copy.c -o file_copy
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ ./file_copy
Enter source file name: gcc file_copy.c -o file_copy^Z
[1]+  Stopped                  ./file_copy
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ ./file_copy
Enter source file name:      File1.txt
Enter destination file name: File2.txt
Error opening source file: No such file or directory
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ ./file_copy
Enter source file name: File.txt
Enter destination file name: File2.txt
Error opening source file: No such file or directory
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ nano file_copy.c
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ gcc file_copy.c -o file_copy
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ echo "This is my OS Lab Experiment 2." > File1.txt
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ cat File1.txt
This is my OS Lab Experiment 2.
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ ./file_copy
Enter source file name: File1.txt
Enter destination file name: File2.txt

File copied successfully.
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ cat File2.txt
This is my OS Lab Experiment 2.
```

```

bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ ./file_copy
Enter source file name: File1.txt
Enter destination file name: File2.txt

File copied successfully.
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ cat File2.txt
This is my OS Lab Experiment 2.
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ nano process_states.c
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ gcc process_states.c -o process_states
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ ./process_states
Creating Child Process...

===== PARENT PROCESS =====
Parent PID      : 1246
Child PID       : 1247
Process State   : Waiting
===== CHILD PROCESS =====
Child PID       : 1247
Parent PID (PPID) : 1246
Process State   : Running
Process State   : Terminated
Process State   : Running
Child Process Completed.
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ nano wait_example.c
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ gcc wait_example.c -o wait_example
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ ./wait_example
Parent Process PID : 1282

Child 1 Created
Child 2 Created
PID : 1283
PID : 1284
PPID : 1282
PPID : 1282
Parent waiting for all child processes...

Child 3 Created
PID : 1285
PPID : 1282
Child 1 Finished

One child process completed.
Child 2 Finished

One child process completed.
Child 3 Finished

All Child Processes Completed.
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ nano waitpid_example.c
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ gcc waitpid_example.c -o waitpid_example
bharath@DESKTOP-6D34C2R:~/Experiment-1/Support$ ./waitpid_example
Parent PID : 1319

Child 1
PID : 1320
PPID : 1319
Child 2
PID : 1321
PPID : 1319
Parent waiting using waitpid()

Child 3
PID : 1322
PPID : 1319
Child 1 Completed

Child with PID 1320 terminated.
Child 2 Completed

Child with PID 1321 terminated.
Child 3 Completed

Child with PID 1322 terminated.

All Child Processes Finished.

```